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Evaluation of the effect of empagliflozin on prevention of atrial fibrillation after coronary artery bypass grafting: a double-blind,

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This study is aimed at evaluating the effect of empagliflozin in preventing atrial fibrillation after coronary artery bypass grafting (CABG). Eighty-two patients who fulfilled the inclusion criteria were allocated to the empagliflozin group (43) (39). In two groups, patients received empagliflozin or placebo tablets 3 days before surgery and on the first three postoperative days (for 6 days) in addition to the standard regimen during hospitalization. During the first 3 days after surgery, types of arrhythmias after cardiac surgery, including supraventricular arrhythmias, especially postoperative atrial fibrillation (POAF), ventricular arrhythmias, and heart blocks, were assessed by electrocardiogram monitoring. in the treatment group was lower compared to the control group; however, this reduction was statistically non-significant (0.09). The frequency of ventricular tachycardia was reduced significantly in the treatment group versus patients in the (0.02). Also, a significant reduction in the frequency of premature ventricular contractions (PVCs) was seen in (0.001). After the intervention, CRP levels were significantly less in the empagliflozin group compared to the control group in the third postoperative day (0.04). of empagliflozin effectively reduced the incidence of ventricular arrhythmia in patients undergoing CABG surgery.

SGLT2 inhibitor · Empagliflozin · Atrial fibrillation · Ventricular arrhythmia · Coronary artery bypass grafting

 bypass grafting (CABG) and represent a significant cause rhythmias are less prevalent (Peretto et al. 2014).

Postoperative atrial fibrillation (POAF) is a common 15–40%

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(ICU) (Abbaszadeh et al. 2012; Bohatch Júnior et al. 2015; Gamst et al. 2015). 1–5 days after surgery, peaks from 24 to 72 h, and decreases within 4 to 7 days after surgery (Silva et al. 2010 et al. 2014). sinus rhythm (Auer et al. 2005; Fuster et al. 2006 et al. 2013; Bohatch Júnior et al. 2015).

dia (VT) and ventricular fibrillation (VF) are rare in post- lar function (Pires et al. 1995; Mouws et al. 2017).

intraoperative atrial ischemia, inflammation (C-reactive pro imbalance (Angelini et al. 1987 1997; Stamou et al. 2000; Tomic et al. 2005). postoperative arrhythmias (Mathew et al. 2004 et al. 2005; Peretto et al. 2014; Ismail et al. 2017; Seo et al. 2021).

used as the first-line agent for patients with HF (McDonagh 2021; Heidenreich et al. 2022).

shown that empagliflozin could have positive effects in the control of cardiac arrhythmias (Azam et al. 2021 2021; Silva Dos Santos et al. 2023). include the anti-inflammatory effect (Hattori 2018) oxidant effect (Watanabe and Singh 2020) the endothelium's normal function (Mone et al. 2022) load (Baker et al. 2014) (INa) (Philippaert et al. 2021) overload in myocytes (Kadosaka et al. 2023) et al. 2022; Wu et al. 2022). Accordingly, empagliflozin may be effective in arrhythmias prevention after CABG, such as atrial fibrillation (AF) and ventricular arrhythmias.

So far, no studies have been performed on the effects of empagliflozin in preventing AF and ventricular arrhythmias able role of empagliflozin in reducing and preventing POAF .

Materials and methods

2022 2023. (: 20120520009801 9) 1401.129). confidentiality is ensured. All patients gave written informed

Eighty-two patients who fulfilled the following inclusion ing elective off-pump CABG, aged between 18 and 70 years old, and using aspirin, statins, β-blockers, and angiotensin- ( ) ( ) 30) 20 ml/min/1.73 ^{2}) insufficiency (increased liver enzymes more than three times flozin or other SGLT2 inhibitor drugs during the 2 weeks ( ) (18.5 kg/m^2) .

Randomization and blinding

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. Placebo and empagliflozin were filled in bottles and numbered (between 1 and 82) by . The boxes were delivered to the patients who fulfilled the tained six tablets, which was enough for 6 days. The cardi SPSS file by the pharmacy technician based on the allocation defined A and B.

ACE inhibitor or ARB, β blocker, statin, and aspirin) were flozin) group. Tablets containing 10 mg empagliflozin with Patients in the treatment group received one empagliflozin 10 mg tablet once daily from 3 days before surgery and on the first three postoperative days. Individuals in the control same shape, size, and color of empagliflozin tablets.

for a minimum of 72 h, and the cardiology resident should read ECG every 12 h. Supraventricular arrhythmias, espe

aspirin, a β-blockers, a statin, and an ACE inhibitor or

the inflammatory process in humans. CRP is an acute- in response to acute and chronic inflammation. ELISA and profile preoperatively and on the 3rd day after the surgery.

baseline and 5–7 days after surgery. It was utilized to

sified as a volume index greater than 34 ml/m^2 and 29 ml/m^2 in women. RA enlargement was defined as volume index greater than 23 ml/m^2 in men and 19 ml/m^2 in women (Baysan et al. 2017).

evaluated in the first 72 h. POAF is characterized by at least one episode of AF that lasts more than 5 min, and (AVRT), and atrial flutter in the first 72 h; (2) the inci (PVC), and ventricular fibrillation (VF) in the first 72 h; (3) (4) (5)

ing CABG (Zipes et al. 2019). and 5% type I error detects an absolute difference of 25% in

... 24. ... 56.97 ± 7.32 years were included in our study. The CONSORT flow diagram is shown in Fig. 1. ... parameters of the two groups are presented in Table 1. There were no statistically significant differences between ... ($p > 0.05$).

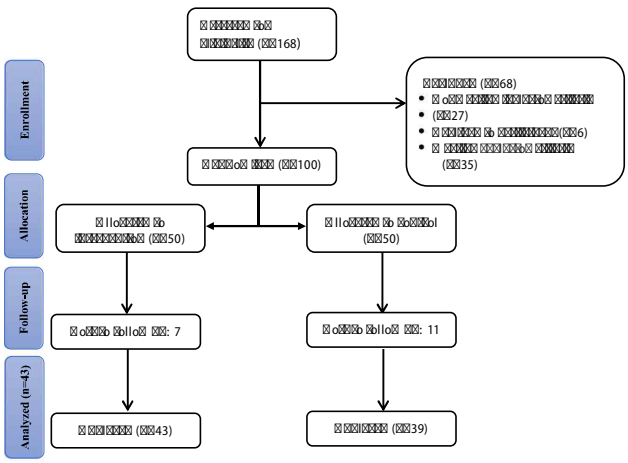

Fig. 1 CONSORT flow chart for trial recruitment

Comparison of the frequencies of cardiac arrhythmias between the treatment and placebo

Along with the intervention, ECG monitoring 72 h after the ... in the empagliflozin group in comparison with 11 patients (28.20%) ... ($p = 0.09$). ... incidence of POAF decreased, it was not significantly lower in the empagliflozin group. Ventricular tachycardia was observed in 2 patients (4.65%) in the empagliflozin group ... (23.07%) ... ($p = 0.02$). ... in 12 patients (27.90%) in the empagliflozin group versus 26 ... (66.6%) ... ($p = 0.001$). ... result, empagliflozin significantly reduced the incidence of ...

... ($p = 0.25$) ... ($p = 0.37$) ... 2 (4.65%) ... 5 (11.62%) ... 5 (11.68%) ... 8 (20.52%) patients in the placebo group. Accordingly, no significant differences in the incidence of supraventricular arrhythmias were observed between the groups (Table 2).

... 5 ... (11.62%) in the empagliflozin group and in 7 (17.95%) ... ($p = 0.54$). ... (IV) ... degree 1 was found in 4 patients (9.3%) in the empagliflozin ... (7.69%) ... ($p = 0.54$). ... significant differences in any post-operative branch bundle blocks and AV block (Table 2).

Comparison of electrocardiogram parameters between the treatment and placebo groups

ECG studies revealed statistically significant differences in ... 1 ($p = 0.02$) and 2 days ($p = 0.01$) ... day after surgery were insignificant ($p = 0.12$). ... cally significant on 1, 2, and 3 days after surgery (Table 3).

... and T-wave inversion are not statistically significant between the treatment group and placebo (Table 3).

Comparison of echocardiographic parameters between the treatment and placebo groups

... flozin and the placebo group post-CABG were compared.

Table 1 Baseline characteristics and laboratory findings of patients with acute myocardial infarction

	Empagliflozin (n = 43)	Placebo (n = 39)	p-value		Empagliflozin (n = 43)	Placebo (n = 39)	p-value
Demographics				Laboratory findings at admission			
Age (year)	57.25 ± 7.54	56.69 ± 7.23	0.25	WBC	46.88 ± 8.90	49.74 ± 8.39	0.28
Male gender	22(51.62%)	28(71.79%)	0.07	WBC	11(26.45%)	10(27.02%)	0.77
BMI (kg/m2)	27.17 ± 4.51	26.18 ± 6.05	0.40	WBC	4(25.57%)	5(28.38%)	0.75
Comorbidities				WBC	3(6.98%)	0 (0%)	0.24
Smoking or Tobacco	5(11.63%)	9(12.82%)	0.24	[illegible]	7(16.66%)	8(21.05%)	0.32
Alcohol	2(4.65%)	2(5.13%)	1.00	[illegible]	1(2.43%)	2(5.55%)	0.59
Opioid	7(16.28%)	7(17.95%)	1.00	[illegible]	36(85.7%)	34(87.18%)	0.39
Medical History				[illegible]	15(35.7%)	16(41.01%)	0.61
IHD	27(62.79%)	24(61.52%)	1.00	[illegible]	35(83.33)	34(87.25%)	0.53
MI	4(9.30%)	4(10.26%)	1.00	[illegible]	8(19.04%)	10(25.64%)	0.59
DM	16(51.16%)	9(23.08%)	0.16	[illegible]	26(60.46%)	15(38.46%)	0.076
DLP	21(48.83%)	18(46.15%)	0.82	[illegible]			
HTN	27(62.79%)	24(61.54)	1.00				
Hypothyroidism	2(4.65%)	2(5.13%)	1.00				
Medications				Laboratory findings at discharge			
Antiplatelet	34(79.07%)	26(66.66%)	0.22	[illegible]	124.51 ± 26.72	129.13 ± 14.88	0.34
ACEI	13(30.23)	11(28.20)	1.00	[illegible]	77.44 ± 14.89	76.67 ± 11.70	0.79
ARB	15(34.88%)	16(41.02%)	0.65	[illegible]	73.27 ± 14.89	71.41 ± 15.89	0.58
β-blocker	30(69.77%)	28(71.79%)	1.00	[illegible]	20.86 ± 14.70	19.69 ± 11.93	0.69
Statin	30(69.77%)	31(71.49%)	0.44	[illegible]			
CCB	14(32.56%)	11(28.20%)	0.81	[illegible]	7.90 ± 2.42	7.41 ± 2.04	0.82
[illegible]				[illegible]	13.62 ± 2.07	17.53 ± 2.21	0.10
ST Elevation	10(23.25%)	5(14.70%)	0.26	[illegible]	238.53 ± 70.13	217.49 ± 66.64	0.49
ST Depression	6(13.95%)	6(15.38%)	1.00	[illegible]	27.87 ± 20.74	26.84 ± 13.47	0.07
T Invert	9(20.93%)	12(30.77%)	0.325	[illegible]	35.31 ± 14.00	29.97 ± 14.91	0.07
RBBB	2(4.65%)	1(2.56%)	0.10	[illegible]	124.73 ± 37.51	124.08 ± 49.71	0.40
LBBB	5(11.63%)	0(0%)	0.36	[illegible]	145.67 ± 68.32	142.91 ± 79.13	0.38
Second- degree AV [illegible]	1(2.32%)	1(2.56%)	1.00	[illegible]	65.86 ± 28.21	64.24 ± 34.19	0.72
PR interval	0.12 ± 0.0.2	0.13 ± 0.03	0.12	[illegible]	33.97 ± 11.85	34.60 ± 10.88	0.31
QRS interval	0.08 ± 0.11	0.054 ± 0.014	0.20	[illegible] (BUN/Cr)	34.50 ± 11.63	28.77 ± 8.69	0.17
QT interval	0.41 ± 0.08	0.42 ± 0.043	0.51	[illegible]	1.14 ± 0.30	1.12 ± 0.23	0.44
				[illegible]	6.73 ± 2.00	6.15 ± 2.03	0.86
				[illegible]	157.67 ± 73.57	148.13 ± 86.14	0.89

[illegible]

[illegible] (Continued)

[illegible]

[illegible]

[illegible]

[illegible] 2 [illegible]

	Empagliflozin	[illegible]	[illegible]
Atrial fibrillation	5(11.62)	11(28.20)	0.09
V[illegible]	2(4.65)	9(23.07)	0.02
[illegible] V[illegible]	12(27.90)	26(66.6)	0.001
[illegible]	2(4.65)	5(12.82)	0.25
[illegible]	5(11.62)	8(20.52)	0.37
[illegible]	5(11.62)	7(17.95)	0.54
[illegible]	1(2.32)	0(0)	1.00
[illegible] 1	4(9.30)	3(7.69)	1.00

[illegible] ([illegible])

[illegible]

[illegible] V [illegible] 43.63 [illegible] 9.56 [illegible] 46.63 [illegible] 9.78 [illegible] statistically significant ([illegible] 0.24). [illegible] graphic finding such as right ventricular systolic dysfunc[illegible] ([illegible] 0.69) [illegible] ([illegible] 0.15) [illegible] ([illegible] 0.64) [illegible] ([illegible] 0.53) [illegible] ([illegible] 0.38) [illegible] ([illegible] 0.77) [illegible] ([illegible] 0.08) showed no signifi[illegible]cant differences between the groups (Table 4).

Comparison of CRP level between the treatment and placebo groups

As shown in Table 3, there was no significant difference in [illegible] ([illegible] 0.96). [illegible] icantly in both empagliflozin (4.96 [illegible] 5.17 [illegible] 138.13 [illegible] 79.32 [illegible] 0.001) [illegible] (4.90 [illegible] 6.49 [illegible] 169.21 [illegible] 55.54 [illegible] 0.001) [illegible]. [illegible] els were significantly lower in the empagliflozin group than [illegible] ([illegible] 0.04) (Table 5).

[illegible] 3 [illegible]

	Empagliflozin	[illegible]	[illegible]
[illegible] 1 [illegible]	13(30.23)	8(20.51)	0.39
[illegible] 2 [illegible]	13(30.23)	5(12.89)	0.06
[illegible] 3 [illegible]	2(4.65)	4(5.13)	1.00
[illegible] 1 [illegible]	16(37.21)	14(35.89)	1.00
[illegible] 2 [illegible]	10(23.25)	5(12.39)	0.26
[illegible] 3 [illegible]	3(6.98)	4(10.26)	0.73
[illegible] 1 [illegible]	20(46.51)	20(51.28)	0.82
[illegible] 2 [illegible]	19(44.18)	16(41.02)	0.83
[illegible] 3 [illegible]	18(41.86)	16(41.02)	1.00
[illegible] 1 [illegible] ([illegible])	0.16 (0.12 0.24)	0.16 (0.12 .024)	0.26 [illegible] 1.14
[illegible] 2 [illegible]	0.16 (0.08 0.24)	0.16 (0.10 0.20)	0.46 [illegible] 0.74
[illegible] 3 [illegible]	0.12 (0.08 0.24)	0.12 (0.10 0.24)	0.86 [illegible] 0.20
[illegible] 1 [illegible] ([illegible])	0.08 (0.04 0.16)	0.08 (0.1 0.2)	0.94 [illegible] 0.07
[illegible] 2 [illegible]	0.08 (0.04 0.12)	0.08 (0.04 0.0.2)	0.63 [illegible] 0.47
[illegible] 3 [illegible]	0.08 (0.04 0.12)	0.08 (0.04 0.16)	0.81 [illegible] 0.23
[illegible] 1 [illegible] ([illegible])	0.39 (0.36 0.5)	0.4 (0.36 .45)	0.02 [illegible] 2.35
[illegible] 2 [illegible]	0.4 (0.35 0.42)	0.4 (0.39 0.45)	0.0.1 [illegible] 2.70
[illegible] 3 [illegible]	0.4 (0.35 0.42)	0.4 (0.36 0.42)	0.12 [illegible] 1.55

[illegible] ([illegible])

[illegible] ([illegible])

[illegible]

[illegible]

Table 4 Comparison of the hemodynamic parameters and clinical outcome between the treatment and placebo groups

	Empagliflozin	Placebo	p-value
Vasoactive drug	43.63 ± 9.56[Mean ± SD]	46.63 ± 9.78[Mean ± SD]	0.24[a]
Vasoactive drug	8(20%)[n (%)]	8(21.05%)[n (%)]	0.69[a]
Inotropic drug	21(51.22%)[n (%)]	14(37.84%)[n (%)]	0.15[a]
Inotropic drug	3(7.5%)[n (%)]	4(10.53)[n (%)]	0.64[a]
Ventilation	40(97.56%)[n (%)]	35(92.10%)[n (%)]	0.53[a]
Ventilation	19(46.34%)[n (%)]	24(63.15%)[n (%)]	0.38[a]
Ventilation	28(68.29%)[n (%)]	25(65.79%)[n (%)]	0.77[a]
Ventilation	13(31.70%)[n (%)]	5(13.15%)[n (%)]	0.08[a]
Intensive care unit stay	31(75.61%)[n (%)]	23(60.52%)[n (%)]	0.15[a]

Vasoactive drug: Use of vasoactive drug during the first 24 h after surgery; Inotropic drug: Use of inotropic drug during the first 24 h after surgery; Ventilation: Use of ventilation during the first 24 h after surgery; Intensive care unit stay: Stay in intensive care unit after surgery

[a]: Chi-square test for categorical variables; [b]: Mann–Whitney U test for continuous variables

Table 5 Comparison of the urine output (UO) between the treatment and placebo groups

Time	UO (ml/kg/h)		p-value
	Empagliflozin	Placebo	
UO in the first 24 h after surgery	4.96 ± 5.17[Mean ± SD]	4.90 ± 6.49[Mean ± SD]	0.96[a]
UO in the first 48 h after surgery	138.13 ± 79.23[Mean ± SD]	169.21 ± 55.54[Mean ± SD]	0.04[a]
Diff	129.95 ± 82.24[Mean ± SD]	151.29 ± 71.03[Mean ± SD]	0.21[a]
p-value	< 0.001	< 0.001	

Diff: UO in the first 48 h after surgery − UO in the first 24 h after surgery

[a]: Chi-square test for categorical variables; [b]: Mann–Whitney U test for continuous variables

Comparison of the hemodynamic parameters and clinical outcome between the treatment and placebo groups

Our results showed that no significant difference in 24-h urine output (UO) after CABG between the empagliflozin and placebo groups (p = 0.96). However, the UO in the first 48 h after CABG was significantly higher in the empagliflozin group than in the placebo group (p = 0.04). Also, the UO in the first 48 h after CABG was significantly higher than the UO in the first 24 h after CABG in the empagliflozin group (p < 0.001). However, no significant difference was found between the UO in the first 48 h after CABG and the UO in the first 24 h after CABG in the placebo group (p = 0.21). The UO in the first 48 h after CABG was significantly higher than the UO in the first 24 h after CABG in the placebo group (p < 0.001). The difference between the UO in the first 48 h after CABG and the UO in the first 24 h after CABG was not significantly different between the empagliflozin and placebo groups (p = 0.21).

The urine output was not significantly different between the empagliflozin group and the placebo group on the 1st day (p = 0.97) after the surgery and on the 2nd day (p = 0.89) after the surgery.

The incidence of postoperative atrial fibrillation (POAF) after CABG (p = 0.16) and the incidence of postoperative ventricular arrhythmias (p = 0.87) were not significantly different between the empagliflozin group and the placebo group. The incidence of postoperative inotropic support (p = 0.3) was not significantly different between the allocation groups (Table 6).

Comparison of laboratory parameters between the treatment and placebo groups

The effect of empagliflozin and placebo on different laboratory parameters is shown in Table 6. Our findings showed that platelet count was significantly higher in the treatment group than in the placebo group on day 1 (p = 0.04), 2 (p = 0.04), and 3 (p = 0.03) after the surgery.

The mean hemoglobin level on day 1 after the surgery in the empagliflozin group (36.63 ± 12.99) was significantly higher than in the placebo group (31.15 ± 10.70) (p = 0.04). The postoperative hemoglobin level on day 2 after the surgery in the empagliflozin group (mean ± SD) was higher than the hemoglobin level on day 1 (p < 0.001). The postoperative hemoglobin level on day 3 after the surgery (mean ± SD) was higher than the hemoglobin level on day 2 (p < 0.001). Regarding the liver enzymes (SGOT), and serum creatinine, no significant differences were detected between the empagliflozin and placebo groups in the 1st to 3rd days after surgery (Table 7).

Discussion

In this study, we investigated the effect of short-term prophylactic use with oral empagliflozin from 3 days before the surgery and on the first three post-CABG days. Empagliflozin was safe and well-tolerated after CABG but not to significant statistics. Treatment with empagliflozin significantly lowered the incidence of ventricular tachycardia and premature ventricular contractions, but it did not affect other cardiac arrhythmias. Inflammation (as measured by white blood cell (WBC) count) was significantly lower in the empagliflozin group than in the placebo group.

The mean hospital stay was significantly shorter on the 1st and 2nd day after the surgery in the empagliflozin group than in the placebo group. Empagliflozin treatment with empagliflozin was the only significant predictor of postoperative hospital stay. The results are summarized in Fig. 2.

Despite the fact that in 2022 the European Society of Cardiology guidelines on the management of patients undergoing cardiac surgery recommended the use of beta-blockers to prevent POAF (Joglar et al. 2024), they are not always of high efficacy. On the positive side, they do not cause major adverse effects. In recent years, research has been carried out in the field of pharmacological prophylaxis of POAF, but the results are not always unambiguous.

Fig. 2 Summary of the results of the 2 treatment (Tsushima et al. 2021). Previous studies further showed that empagliflozin and dapagliflozin have favorable effects on cardiovascular outcomes irrespective of diabetes status (McMurray et al. 2019; Packer et al. 2020). Despite the fact that the

Table 6 [illegible]

	Empagliflozin	[illegible]	
[illegible] (%)	119.34 ± 18.76[illegible]	124.94 ± 8.84[illegible]	0.09[illegible]
[illegible]	122.74 ± 10.45[illegible]	120.58 ± 13.93[illegible]	0.42[illegible]
[illegible]	125.28 ± 11.27[illegible]	124.37 ± 16.34[illegible]	0.77[illegible]
[illegible] (%)	69.66 ± 6.12[illegible]	71.51 ± 5.92[illegible]	0.21[illegible]
[illegible]	71.98 ± 10.26[illegible]	73.20 ± 9.09[illegible]	0.57[illegible]
[illegible]	75.64 ± 9.31[illegible]	76.92 ± 11.01[illegible]	0.57[illegible]
[illegible] (%)	90.39 ± 14.06[illegible]	86.28 ± 10.99[illegible]	0.15[illegible]
[illegible]	87.36 ± 14.19[illegible]	88.27 ± 12.04[illegible]	0.76[illegible]
[illegible]	89.81 ± 12.93[illegible]	88.20 ± 16.45[illegible]	0.62[illegible]
[illegible] (%)	16.44 ± 2.15[illegible]	17.07 ± 1.85[illegible]	0.16[illegible]
[illegible]	21.28 ± 10.35[illegible]	18.56 ± 2.84[illegible]	0.12[illegible]
[illegible]	18.45 ± 3.75[illegible]	19.45 ± 4.53[illegible]	0.30[illegible]
[illegible]	1.64 ± 2.08 [illegible]	1.57 ± 2.82 [illegible]	0.74[illegible]
[illegible]	3.20 ± 0.85[illegible]	3.20 ± 0.6[illegible]	0.97[illegible]
[illegible]	3.20 ± 0.55[illegible]	3.27 ± 0.36[illegible]	0.89[illegible]
[illegible] (ms)	232.2 ± 43.2[illegible]	230.0 ± 42[illegible]	0.87[illegible]
[illegible]	2(1–4) [illegible]	2(1–5) [illegible]	0.16[illegible]
[illegible]	9(5–17) [illegible]	8(5–25) [illegible]	0.30[illegible]
[illegible] (%)	7(4–20) [illegible]	7(2–17) [illegible]	0.87[illegible]
[illegible]	3(6.97%) [illegible]	2(5.12%) [illegible]	0.78[illegible]

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[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

HF and cardiovascular death (Anker et al. 2021[illegible] et al. 2021). [illegible] 2 [illegible] significantly decrease the burden of AF. This finding is sup[illegible] et al. 2020; Fernandes et al. 2021). [illegible] arrhythmias, and sudden cardiac death (Chen et al. 2020[illegible] Zelniker et al. 2020; Fernandes et al. 2021; Gao et al. 2022).

Silva dos Santos et al. demonstrated that empagliflozin [illegible] et al. 2023). [illegible] antiarrhythmic effects of SGLT2 inhibitors, specifically dapagliflozin, on atrial cardiomyocytes. They discovered that dapagliflozin modifies atrial cardiomyocytes' excitability by directly suppressing peak sodium currents (Paasche et al. 2024). [illegible] show any changes (Bode et al. 2021). [illegible] fibrillation (AF) indicates that canagliflozin treatment miti[illegible] with SGLT2 inhibitors, reductions in atrial effective refrac[illegible] et al. 2021). In an animal study, the antiarrhythmic effects of empagliflozin were investigated in a model of global ischemia–reperfusion in rabbits. Empagliflozin treatment was associated with significant improvements in cardiac [illegible] dysregulation (Azam et al. 2021). [illegible] 58 trial, the effects of dapagliflozin were compared with a

	Empagliflozin		
1	36.63 ± 12.99	31.15 ± 10.70	0.04
2	39.51 ± 16.32	34.49 ± 14.50	0.15
3	45.40 ± 17.40	39.18 ± 19.41	0.13
1	1.10 ± 0.29	1.11 ± 0.27	0.93
2	1.13 ± 0.35	1.05 ± 0.31	0.30
3	1.11 ± 0.29	1.17 ± 0.62	0.55
1	146.62 ± 41.51	154.41 ± 50.11	0.45
2	127.00 ± 34.04	130.28 ± 42.84	0.70
3	137.67 ± 45.70	149.13 ± 52.64	0.31
1	58.56 ± 16.67	44.52 ± 12.64	0.19
2	44.52 ± 22.61	41.86 ± 13.33	0.73
3	31.33 ± 16.43	36.92 ± 11.32	0.19
1	34.53 ± 17.56	30.24 ± 17.74	0.40
2	25.94 ± 11.23	23.81 ± 12.5	0.5
3	24.63 ± 10.61	23.66 ± 10.36	0.78
1	13.52 ± 2.93	12.96 ± 2.99	0.38
2	13.14 ± 3.93	12.66 ± 3.69	0.58
3	10.64 ± 2.70	10.05 ± 3.04	0.36
1	214.33 ± 56.80	183.87 ± 42.48	0.04
2	198.12 ± 76.73	168.72 ± 34.50	0.04
3	226.14 ± 46.5	183.32 ± 50.1	0.03

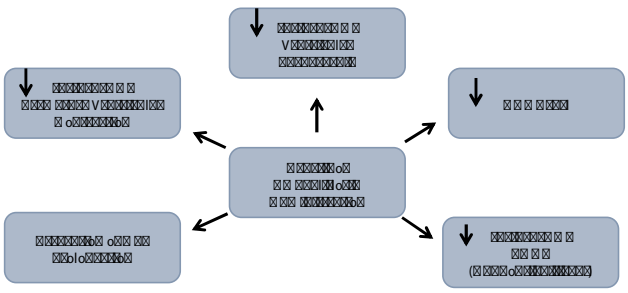

Fig. 2 Beneficial of empagliflozin administration in the study

placebo in DM2 patients. It was found that dapagliflozin reduces the incidence of AF and atrial flutter (AFL) compared to the placebo group (Zelniker et al. 2020). In DAPA-HF trial, the effects of dapagliflozin were found that adding dapagliflozin at a dose of 10 mg daily to (Fernandes et al. 2021).

The precise mechanisms of these antiarrhythmic effects studies have suggested that these effects are mediated directly through attenuation of inflammation, oxidative and suppression of fibrosis and endothelial dysfunction (Wu et al. 2022). receptor 3 reduced myocardial inflammation and structural remodeling (Jing et al. 2022). that empagliflozin can significantly inhibit the activity of that treatment with empagliflozin was associated with an anti-inflammatory effect.

Also, our study proposed that empagliflozin prevents QTc prolongation. Studies have suggested that empagliflozin regulation, affected potassium channels, and inhibited the cytes (Baris et al. 2020; Philippaert et al. 2021; Barış et al. 2023). In healthy humans, empagliflozin does not prolong the QT interval (Ring et al. 2013). empagliflozin significantly attenuates QTc prolongation in rats due to sotalol (Özgür Barış et al. 2021). study by Barış VÖ et al., it was found that empagliflozin (10 mg/kg) significantly ameliorated QT and QTc prolongation induced by intoxication amitriptyline (100 mg/kg) (Barış et al. 2021). Xue et al. reported that treatment with empagliflozin reduced ventricular arrhythmia susceptibility post-myocardial-infarction and empagliflozin significantly et al. 2022).

Our study showed that empagliflozin can significantly level as an inflammatory factor. In the case of AF, the

inhibitors could not prevent AF occurrence (Zhang et al. 2023) [illegible]. [illegible] the effects of empagliflozin on the prevention of atrial fibrillation after coronary artery bypass grafting have been investigated for the first time.

[illegible]

[illegible]:

1. [illegible]
2. We did not assess the effects of different doses of empagliflozin on preventing POAF and ventricular arrhythmia.
3. [illegible]
4. The study was not designed for different diabetic and [illegible]
5. [illegible]

[illegible]

The result of this study indicates that empagliflozin treat[illegible] still warranted to examine the safety and impact of different doses of empagliflozin and other possible advantages and [illegible]

[illegible] ACE: Angiotensin-converting enzyme; AF: Atrial fibrillation; AFL: Atrial flutter; AVNRT: Atrioventricular nodal reentrant tachycardia; AVRT: Atrioventricular reentrant tachycardia; ARB: Angiotensin receptor blocker; AR: Aortic regurgitation; AV: Atrioventricular; CABG: Coronary artery bypass grafting; CBC: Complete blood count; COPD: Chronic obstructive pulmonary disease; CPB: Cardiopulmonary bypass; Sr Cr: Serum creatinine; CRP: C[illegible]tein; DBP: Diastolic blood pressure; DM2: Type 2 diabetes mellitus; ECG: Electrocardiography; FBS: Fasting blood sugar; HbA1c: Glycated hemoglobin; HF: Heart failure; HTN: Hypertension; ICU: Intensive care unit; IHD: Ischemic heart disease; iNOS: Inducible nitric oxide synthase; IL-6: Interleukin-6; LBBB: Left bundle branch block; LVEF: Left ventricular ejection fraction; MR: Mitral regurgitation; POAF: Postoperative atrial fibrillation; PLT: Platelet; PR: Pulmonary regurgitation; PVC: Premature ventricular contraction; RR: Respiratory rate; RBBB: Right bundle branch block; RVDD: Right ventricular diastolic dysfunction; TC: Total cholesterol; TG: Triglyceride; TNF-alpha: Tumor necrosis factor-alpha; TR: Tricuspid regurgitation; SBP: Systolic blood pressure; SGLT2: Sodium-glucose transporter 2; SGOT: Serum glutamic-oxaloacetic transaminase; SGPT: Serum glutamic-pyruvic transaminase; VT: Ventricular tachycardia; VF: Ventricular fibrillation; WBC: White blood cell

[illegible]

[illegible] the statistical analysis of the data. Batool Zarei wrote the first draft of [illegible] approved the final manuscript. The authors declare that all data were [illegible].

[illegible]

[illegible]

[illegible]

[illegible] (IR.[illegible].1401.129) [illegible] 201205200098019. [illegible].

[illegible]

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